

Nguyen Khac Long

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Education

- B.Eng. in Automation and Informatics, Specialization in Automation** *Sep. 2021 – Present*
International School, Vietnam National University, Hanoi *Hanoi, Vietnam*
- **GPA:** 3.71/4.00 (**Top 4% of cohort**).
 - **Relevant Coursework:** Control Systems, Robotics, Modeling and Simulation of Control Systems, Artificial Intelligence.
 - **Research Interests:** Nonlinear and adaptive control, learning-based control, intelligent robotics, and autonomous systems.
 - **International Academic Experience (2025–2026):** CCU Research Internship; NUS Semester Exchange; NTU Global Connect Fellowship; NUS ASEAN SPARK Fellowship.

Research Experience

- Undergraduate Researcher** *Sep. 2022 – Present*
Intelligent Control and Artificial Intelligence Laboratory, VNUIS *Hanoi, Vietnam*
- **6-DOF Robotic Manipulator – Nonlinear and Learning-Based Control:** Developing dynamic-surface and adaptive neural controllers for trajectory tracking under uncertain dynamics and external disturbances; analyzing closed-loop stability using Lyapunov methods, validating the design in MATLAB/Simulink, and extending the framework toward Kinova Gen3 Lite experiments. The part of this work led to a first-author manuscript currently under review at RIVF 2026.
 - **AUV – Adaptive Nonlinear Control for Underwater Wireless Power Transfer:** Investigating adaptive nonlinear control for an underwater wireless power transfer (UWPT) system integrated with an AUV, with emphasis on regulating power-transfer dynamics under coupling variations, parameter uncertainties, and changing operating conditions.
 - **3-DOF Parallel Delta Robot – Adaptive Fuzzy Backstepping:** Implemented adaptive fuzzy backstepping with fuzzy approximation of uncertain nonlinear dynamics for trajectory tracking; the work resulted in a peer-reviewed paper presented at ICTA 2024 and published by Springer.
- Research Fellow | Underwater Wireless Power Transfer for AUVs** *Feb. – Mar. 2026*
Water & Energy Research Lab, School of EEE, Nanyang Technological University *Singapore*
- Modeled a relay-coil-enhanced 3D wireless power transfer system using equivalent-circuit analysis and MATLAB; evaluated the effects of coupling, load resistance, and relay-coil resonance on transfer efficiency and output power.
 - Designed a disc-shaped AUV concept in SolidWorks by integrating mechanical, control, underwater communication, docking, and wireless-charging constraints.
 - Presented the project at the NTU Global Connect Fellowship Final Research Presentation and received the **Most Popular Poster Award**.

- Research Intern | Smart Manufacturing and Data-Driven Modeling** *May 2025 – Present*
AISC Lab, AIM-HI, National Chung Cheng University *Taiwan / Remote*
- Built and benchmarked Random Forest, XGBoost, and Support Vector Regression models for cutting-force and tool-wear prediction in CNC milling.
 - Processed 12 onsite experimental datasets and 27 Taguchi-designed datasets through preprocessing, feature analysis, model training, and comparative evaluation for tool-condition monitoring.

Publications & Manuscripts

- **Khac Long Nguyen**, An Nguyen Huu, Nhat Dan Nguyen, Thuy Vo Thi Cam, and Hai Le Xuan, “Event-Triggered Adaptive Neural Dynamic Surface Control with Butterworth Filtering for Six-DOF Manipulator Tracking,” manuscript under review at the *20th International Conference on Computing and Communication Technologies (RIVF)*, 2026.
- A. T. Le, G. L. Truong, H. Q. Nguyen, T. A. Than Ngoc, B. T. The, **L. N. Khac**, et al., “Adaptive Fuzzy Logic in Backstepping Control for 3-DOF Parallel Delta Robot,” in *Advances in Information and Communication Technology: Proceedings of ICTA 2024*, Springer, 2025, pp. 539–546. [doi:10.1007/978-3-031-80943-9_57](https://doi.org/10.1007/978-3-031-80943-9_57).
- D. M. Tuan, T. M. T. Kiet, N. V. Tung, N. X. The, N. A. Chau, P. T. Ngoc, **N. K. Long**, et al., “YOLOv5s Model with Applied Activation Functions for Personal Protective Equipment Detection in Construction Sites,” *Proceedings of the National Conference on Smart Technology Applications in Industry 4.0, Smart City, and Sustainability (STAIS)*, 2023.

Honours and Awards

- **Second Poster Prize**, Science Poster Session, Vietnam Summer School of Science, 2026.
- **Most Popular Poster Award**, NTU Global Connect Fellowship Final Research Presentation, 2026.
- **NTU Global Connect Fellowship**, selected as one of 39 young scholars worldwide, 2026.
- **NUS ASEAN SPARK Fellow**, NUS Enterprise Summer Programme in Entrepreneurship, 2026.
- **Toshiba Scholarship**, TOSHIBA International Foundation, 2024.
- **Fighting Spirit Prize**, IEEE Southeast Asia Circuits and Systems Society Hackathon, 2022.
- **Merit-based Academic Scholarship**, five-semester recipient for excellent academic performance, 2021–2025.

Skills

- **Control & Dynamical Systems:** nonlinear system modeling, adaptive control, robust control, backstepping, dynamic surface control, Lyapunov stability analysis, neural-network-based control.
- **Robotics & Simulation:** MATLAB/Simulink, ROS 2, Gazebo, SolidWorks; Kinova Gen3 Lite, parallel Delta robot, AUV modeling and simulation.
- **Programming & Scientific Computing:** MATLAB, Python (NumPy, Pandas, Matplotlib, scikit-learn, PySINDy), C/C++, LaTeX, Git/GitHub.
- **Data-Driven Methods:** Random Forest, XGBoost, Support Vector Regression, SINDy, SVD.
- **Languages:** Vietnamese (native), English (IELTS 6.5).

References

Asst. Prof. Yun Yang — School of Electrical & Electronic Engineering, Nanyang Technological University
Email: yun.yangntu.edu.sg

Prof. Her-Terng Yau — Director, Advanced Institute of Manufacturing with High-tech Innovations, National Chung Cheng University
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Dr. Le Xuan Hai — Vice Dean, Faculty of Engineering and Technology, International School – Vietnam National University, Hanoi
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